

Limits

Week 2D

Overview

Limits

Key Ideas - Introduction to Limits

- Limits by Substitution*
- Limits by Factorisation*
- Infinite Limits*

Key Idea – Introduction to Limits

Calculus is based on the fundamental concept of **limits**.

A Limit describes how a function $f(x)$ behaves **near** a given point $x = a$ without actually using the value at a . We can usually deduce (but not prove) the value of the limit at a by evaluating $f(x)$ near the value a .

Example:

Let $f(x) = \frac{x^2 - 4}{x - 2}$, $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 2\}$. By using the table of values below, how does $f(x)$ behave near $x = 2$?

$x \rightarrow 2$ from left	$f(x)$	$x \rightarrow 2$ from right	$f(x)$
1	3	3	5
1.5	3.5	2.5	4.5
1.9	3.9	2.1	4.1
1.99	3.99	2.01	4.01
1.999	3.999	2.001	4.001
1.9999	3.9999	2.0001	4.0001
$x \rightarrow 2$	$f(x) \rightarrow 4$	$x \rightarrow 2$	$f(x) \rightarrow 4$

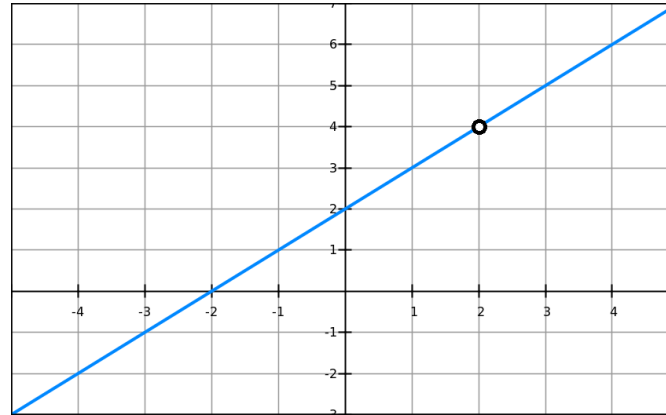
We can see from the table that, as x gets closer to 2 from either side, the value of $f(x)$ gets close to 4.

This is written as:

$$\lim_{x \rightarrow 2} f(x) = 4$$

and we say “the limit of $f(x)$ as x approaches 2 is 4”.

From the graph we can also see that the value of $f(x)$ gets closer to 4 as x gets closer to 2.

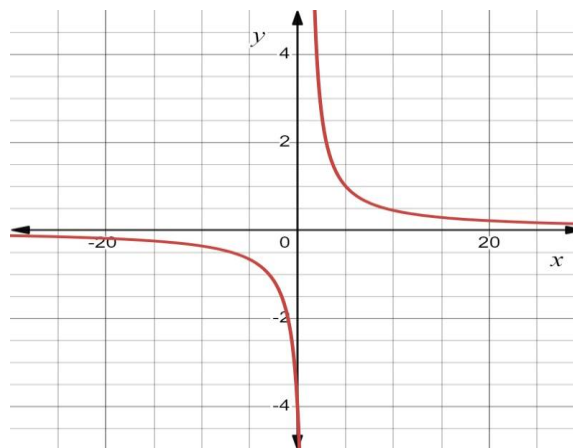


Note: This function has a gap at $x = 2$, as division by zero is undefined. Although $f(x)$ is undefined at this point, there is a limit as x approaches 2. This explains why the domain of $f(x)$ does not include $x = 2$.

Example:

Let $f(x) = \frac{4}{x-1}$, $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 1\}$. The graph of the function is given below.

How does $f(x)$ behave near $x = 1$, i.e. $\lim_{x \rightarrow 1} f(x)$?



From the graph we deduce that there is **no single value that $f(x)$ approaches** when x approaches 1. This is because, as x gets closer to 1 from the left, $f(x)$ approaches $-\infty$ and as x closer to 1 from the right, $f(x)$ approaches $+\infty$.

Therefore, we say **$\lim_{x \rightarrow 1} f(x)$ Does Not Exist (or DNE).**

Note: This function has a gap at $x = 1$, as division by zero is undefined. $f(x)$ is undefined at this point and there is NO limit as x approaches 1. This explains why there is an asymptote at $x = 1$ and why the domain of $f(x)$ does not include $x = 1$.

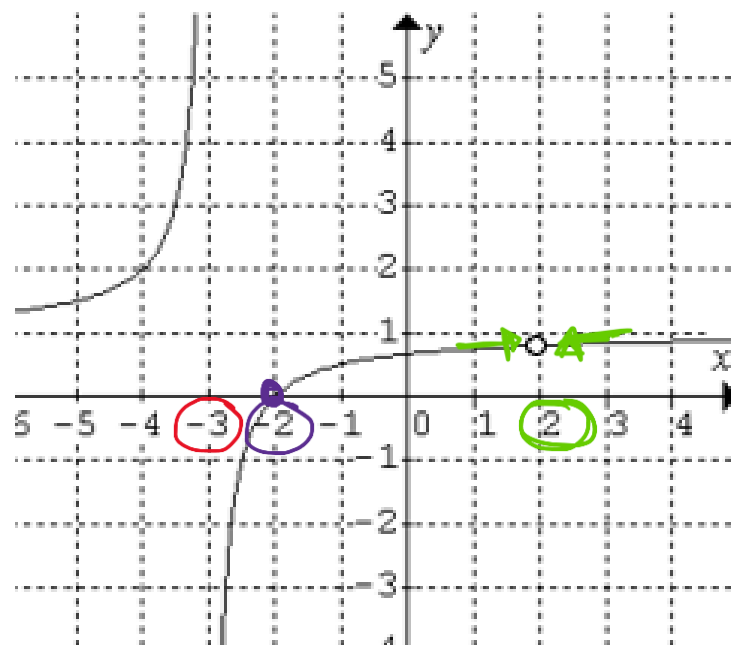
Exercise:

Consider $f(x) = \frac{x^2 - 4}{x^2 + x - 6} = \frac{x^2 - 4}{(x+3)(x-2)}$, $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 2, x \neq -3\}$. By using the graph below, find the following limits:

(a) $\lim_{x \rightarrow -3} f(x) = \text{DNE}$

(b) $\lim_{x \rightarrow 2} f(x) = \bigcirc$

(c) $\lim_{x \rightarrow 2} f(x) = 1.9$



Is the statement $\lim_{x \rightarrow -2} f(x) = f(-2)$ true or false?

false as $f(-2)$ does not exist as there is an open circle at $x = -2$ meaning there is no y value

Informal Definition

The **limit** is a value, say L , that a function $y = f(x)$ approaches as x approaches (gets closer and closer) to a certain value, say a .

In mathematical notation we write this as: $\lim_{x \rightarrow a} f(x) = L$

In other words:

The $\lim_{x \rightarrow a} f(x)$ asks the question:

“What value does $y = f(x)$ get **close to** as x gets **close to** a .”

We are NOT concerned with what happens at $x = a$, ONLY what is happening when x gets “very close to” a .

Key Idea – Limits by Substitution

If the graph of a function is “**unbroken**” at a point $x = a$, (no gaps), then we use **substitution** to evaluate the limit. That is, we merely substitute a into f as in

$$\lim_{x \rightarrow a} f(x) = f(a)$$

If substitution gives a sensible answer (no divisions by 0 or ‘ $\frac{0}{0}$ ’) then that is what we use. Note that a limit of zero is a sensible answer. These limits are easy to evaluate.

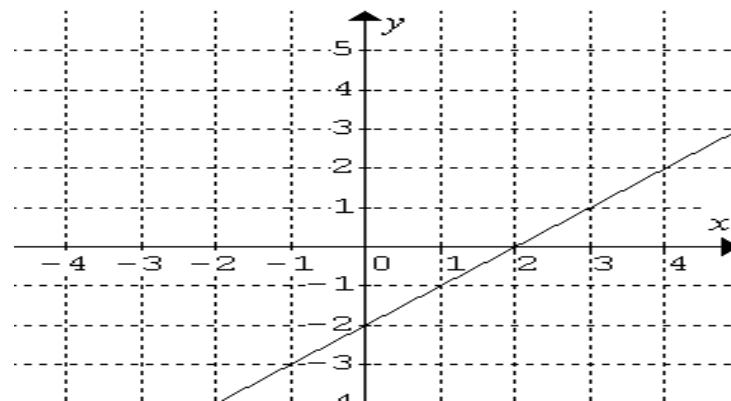
Examples:

Let $f(x) = (x - 2)$. Evaluate the following limits: $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow -1} f(x)$.

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x - 2) = -1 = f(1)$$

$$\lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} (x - 2) = -3 = f(-1)$$

Geometrically, we can also see that the graph of the function is unbroken at the points $x = 1$ and $x = -1$.



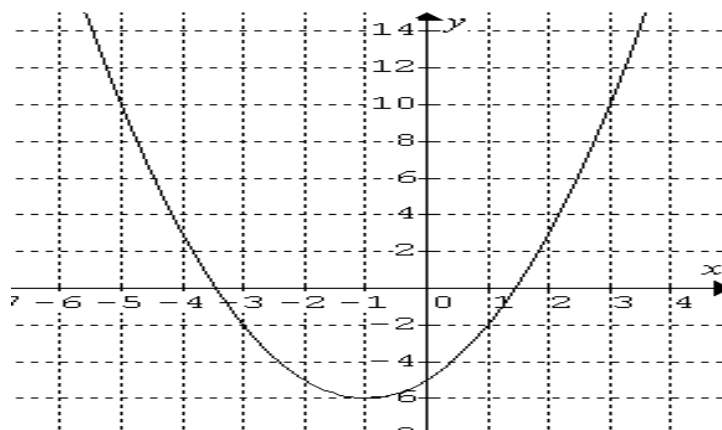
Example:

Let $f(x) = (x^2 + 2x - 5)$. Evaluate $\lim_{x \rightarrow 3} f(x)$.

$$\begin{aligned}\lim_{x \rightarrow 3} (x^2 + 2x - 5) &= 3^2 + 2 \times 3 - 5 \\ &= 10\end{aligned}$$

Geometrically, we can see that the graph of the function is unbroken at the point $x = 3$.

We can also see from the graph that the value of y gets closer to 10 as x gets closer to 3.



Example:

Let $f(x) = \ln x$. Evaluate : $\lim_{x \rightarrow \frac{1}{2}} f(x)$ and $\lim_{x \rightarrow 1} f(x)$.

$$\lim_{x \rightarrow \frac{1}{2}} \ln x = \ln \frac{1}{2}$$

$$= -\ln 2$$

$$= -0.6931$$

$$= f\left(\frac{1}{2}\right)$$

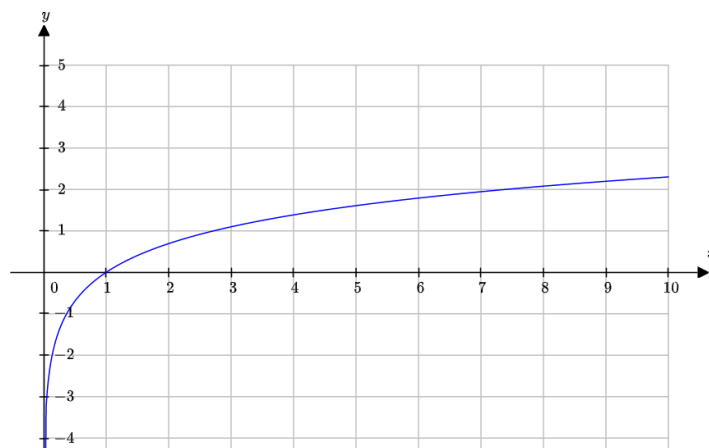
$$\lim_{x \rightarrow 1} (\ln x) = \ln 1$$

$$= 0$$

$$= f(1)$$

Geometrically, we can see that the graph of the function is unbroken at the points $x = \frac{1}{2}$ and $x = 1$.

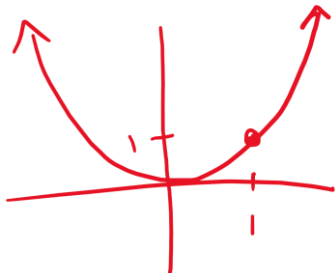
We can also see from the graph that the value of $f(x)$ gets closer to -0.693 as x gets closer to $\frac{1}{2}$ and $f(x)$ gets closer to 0 as x gets closer to 1 .



Exercise:

Make small sketches and find these limits for the following functions.

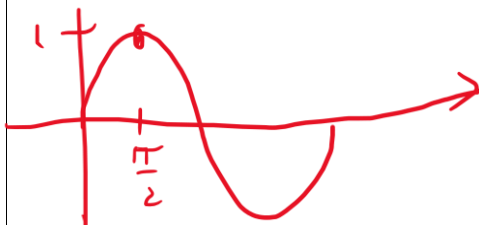
(a) $y = f(x) = x^2$, $\lim_{x \rightarrow 1} f(x) = 1^2 = 1$



(b) $y = g(x) = e^x$, $\lim_{x \rightarrow 2} g(x) = e^2 = 7.389$



(c) $y = h(x) = \sin x$, $\lim_{x \rightarrow \frac{\pi}{2}} h(x) = \sin\left(\frac{\pi}{2}\right) = 1$



Exercise:

Evaluate the following limits by substitution:

$$(a) \quad \lim_{x \rightarrow 0} (x^2 + 3x + 1) = 0^2 + 3 \times 0 + 1 = 1$$

$$(b) \quad \lim_{t \rightarrow 1} (3t - 1)(5t^2 + 2) = (3 \times 1 - 1)(5 \times 1^2 + 2) = 2 \times 7 = 14$$

$$(c) \quad \lim_{x \rightarrow -1} 3(2)^{x+5} = 3 \times 2^{-1+5} = 3 \times 2^4 = 3 \times 16 = 48$$

$$(d) \quad \lim_{x \rightarrow -1} \frac{2x^2 - 2}{x^2 - 3} = \frac{2 \times (-1)^2 - 2}{(-1)^2 - 3} = \frac{0}{-2} = 0$$

$$(c) \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x + \cos x}{\tan x} = \frac{\sin \frac{\pi}{4} + \cos \frac{\pi}{4}}{\tan \frac{\pi}{4}} = \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{1} = \frac{2}{\sqrt{2}}$$

Simplification Techniques

Many **important** limits exist that cannot be found by substitution. These are the **tricky** problems and the ones you are more likely to see.

The difficulty is that the function is usually not in the appropriate fractional “form” for you to deduce the answer and you must use algebra to change the form.

How?

Usually by “simplifying” the function using factorisation.

Key Idea – Limits of the Form $\frac{0}{0}$

Sometimes when we evaluate a limit by direct substitution, we obtain an expression of the form

$$\frac{0}{0}$$

We refer to this as “indeterminant form”, which appears to be undefined.

If this is the case, then we need to

“Reshape or remodel” our given function $f(x)$

without creating a new function.

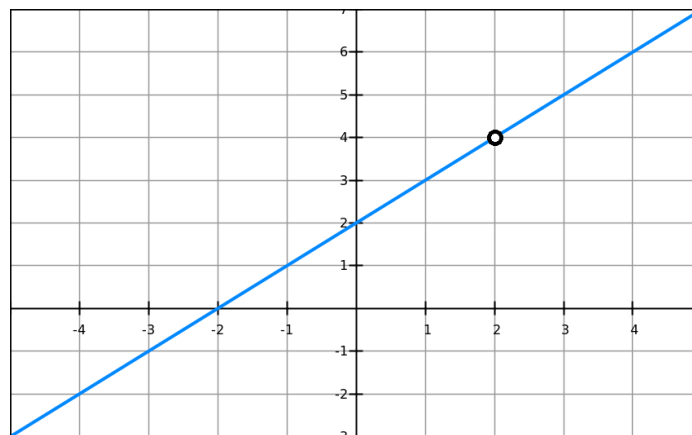
Limits by Factorisation

Factorisation of the numerator and/or denominator and then cancelling common factors is one way to reshape our functions. The limit is then determined by substitution into the simplified expression.

Example: Let $f(x) = \frac{x^2 - 4}{x - 2}$.

What is the domain of this function? $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 2\}$

What is happening graphically for $f(x) = \frac{x^2 - 4}{x - 2}$?



Geometrically we can see that the function has a gap at $x = 2$

Evaluate $\lim_{x \rightarrow 2} f(x)$

When we substitute $x = 2$ we obtain $\frac{0}{0}$ which is in indeterminate form, therefore we need to simplify by factorisation.

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{(x - 2)} \\ &= \lim_{x \rightarrow 2} (x + 2) \\ &= 2 + 2 \\ &= 4\end{aligned}$$

Note: The setting out is an important part of finding limits. Notice that the limit notation is not removed until the substitution step.

Exercise:

Evaluate the following limits:

$$\begin{aligned}\text{(a)} \quad \lim_{x \rightarrow 1} \frac{x^2 - x}{x - 1} &= \lim_{x \rightarrow 1} \frac{\cancel{x(x-1)}}{\cancel{(x-1)}} \\ &= \lim_{x \rightarrow 1} x \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 + x - 6} &= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x+2)}{(x+3)\cancel{(x-2)}} \\ &= \lim_{x \rightarrow 2} \frac{x+2}{x+3} \\ &= \frac{2+2}{2+3} \\ &= \frac{4}{5}\end{aligned}$$

Key Idea – Limits of the Form $\frac{\text{number}}{0}$

Sometimes when we evaluate a limit by direct substitution, we obtain an expression of the form

$$\frac{k}{0}$$

Where k is any number other than zero.

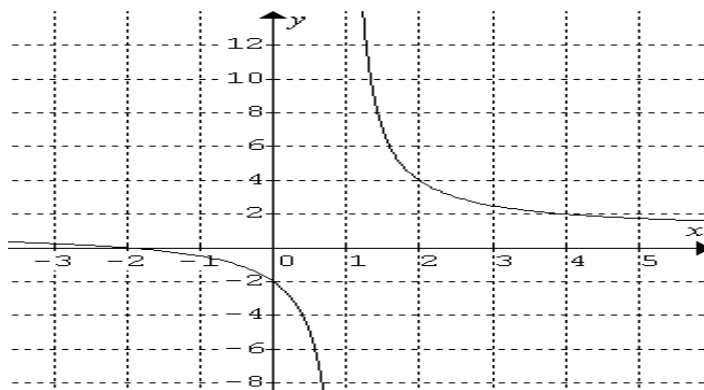
If this is the case we say that

$$\lim_{x \rightarrow a} f(x) \text{ Does Not Exist}$$

Example: Let $f(x) = \frac{x+2}{x-1}$

What is the domain of this function? $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 1\}$

What is happening graphically for $f(x) = \frac{x+2}{x-1}$?



Geometrically we can see that the function has a vertical asymptote at $x = 1$. We can also see that as x gets close to 1, y does not get close to any real number, so the limit does not exist (DNE).

Evaluate $\lim_{x \rightarrow 1} f(x)$

$$\lim_{x \rightarrow 1} \frac{x+2}{x-1} = \frac{1+2}{1-1} \\ = \frac{3}{0}$$

By substitution we get " $\frac{3}{0}$ ", so the limit does not exist (DNE).

Exercise:

Evaluate the following limits:

$$(a) \quad \lim_{x \rightarrow 3} \frac{x^2 - 6}{x - 3} = \frac{3^2 - 6}{3 - 3}$$

$$= \frac{9 - 6}{0}$$

$$= \frac{3}{0}$$

$$= \text{DNE}$$

$$(b) \quad \lim_{x \rightarrow 1} \frac{x^2 + 2x + 2}{x^2 + 3x - 4} = \frac{1^2 + 2 \times 1 + 2}{1^2 + 3 \times 1 - 4}$$

$$= \frac{5}{0}$$

$$= \text{DNE}$$

Key Idea –Infinite Limits

Consider

$$\lim_{x \rightarrow \infty} f(x)$$

The question is now,

“What value does $y = f(x)$ get close to as x gets large in magnitude and positive?”

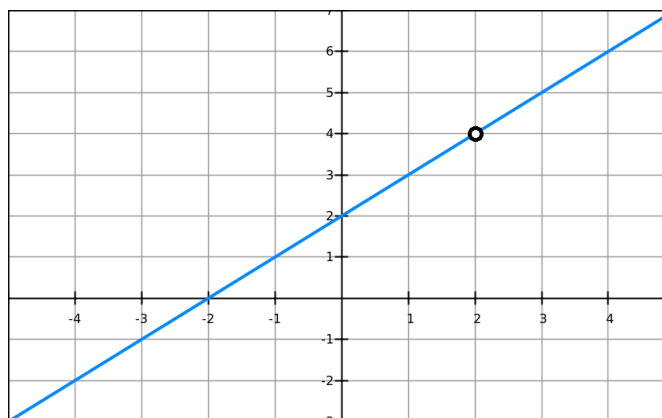
Similarly, we have $\lim_{x \rightarrow -\infty} f(x)$ which asks:

“What value does $y = f(x)$ get close to as x gets large in magnitude and negative?”

Example:

Let $f(x) = \frac{x^2 - 4}{x - 2}$, $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq 2\}$. The graph of the function is given below.

What is $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?



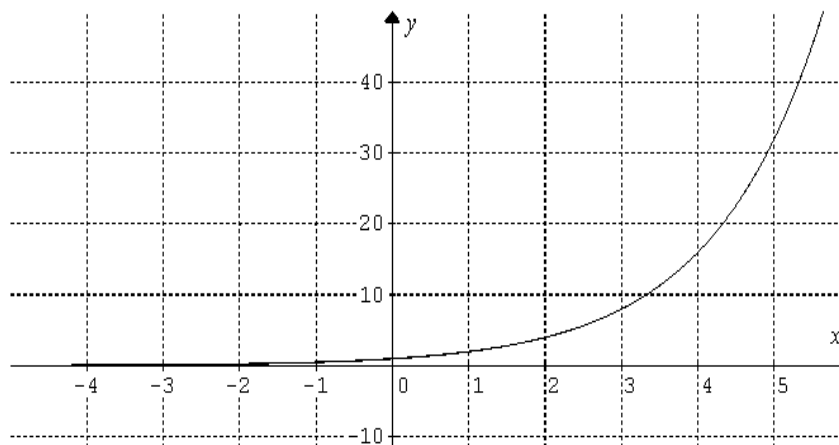
We can see from the graph that, as x approaches ∞ , the value of $f(x)$ approaches ∞ and as x approaches $-\infty$, the value of $f(x)$ approaches $-\infty$. Therefore, we say $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$

Does Not Exist

Example:

Let $f(x) = 2^x$, $\text{Dom } f(x) = \{x \in \mathbb{R}\}$. The graph of the function is given below.

What is $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?



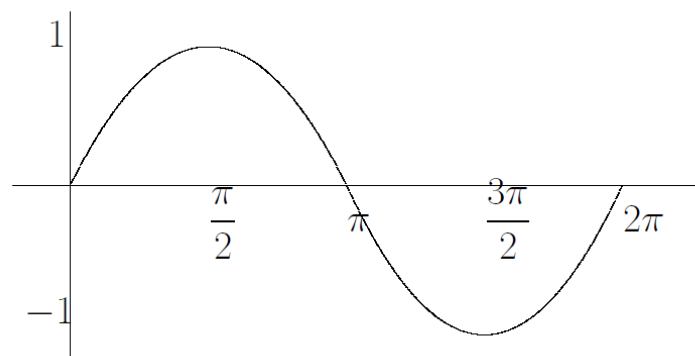
From the graph we deduce that, as x approaches ∞ , the value of $f(x)$ approaches ∞ , therefore, we say $\lim_{x \rightarrow \infty} f(x)$ Does Not Exist. So, $\lim_{x \rightarrow \infty} f(x)$ DNE.

We can also see from the graph that, as x approaches $-\infty$, the value of $f(x)$ approaches 0, therefore, we say $\lim_{x \rightarrow -\infty} f(x) = 0$.

Example:

Let $f(x) = \sin x$, $\text{Dom } f(x) = \{x \in \mathbb{R}\}$. The graph of the function is given below.

What is $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?

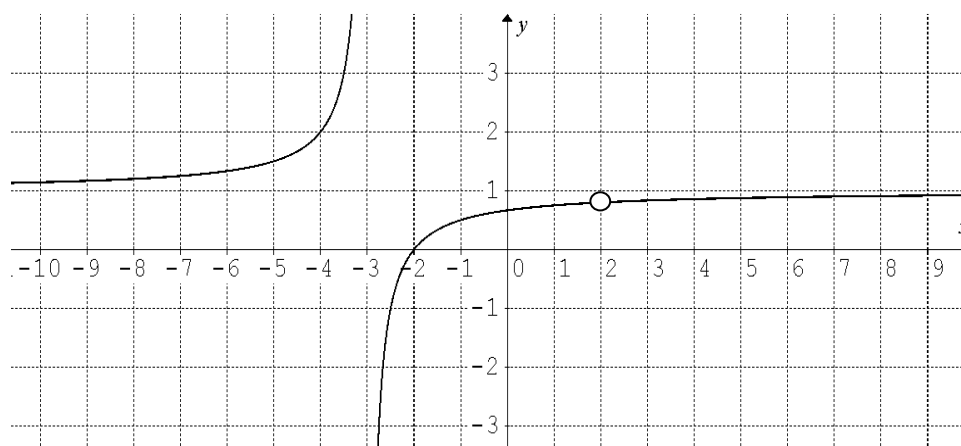


It can be seen from the graph of the $\sin(x)$ function that, $\lim_{x \rightarrow \infty} \sin x$ DNE and $\lim_{x \rightarrow -\infty} \sin x$ DNE

Example:

Let $\lim_{x \rightarrow \infty} \frac{x^2 - 4}{(x + 3)(x - 2)}$, $\text{Dom } f(x) = \{x \in \mathbb{R} : x \neq -3, x \neq 2\}$. By using the graph below find the

following limits: $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$?



As x approaches ∞ (or x approaches $-\infty$), y is approaching the value 1. Therefore,

$$\lim_{x \rightarrow \infty} f(x) = 1 \text{ and } \lim_{x \rightarrow -\infty} f(x) = 1.$$

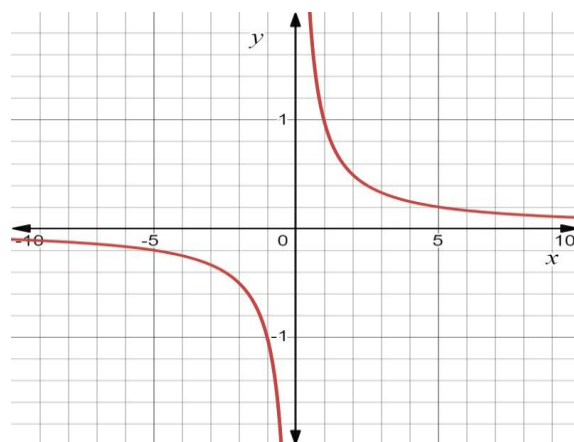
Note: There is a horizontal asymptote at $y = 1$.

Example:

For polynomial functions (e.g. linear functions, quadratic functions, cubic functions etc) there is **no limiting value** for the function as x approaches ∞ and x approaches $-\infty$. The “ends” of the graph are increasing or decreasing without bound.

The graph below involves a special limit that will be encountered in the following examples and exercises.

The graph of the hyperbola $y = f(x) = \frac{1}{x}$.



From this graph we can see that

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 = \lim_{x \rightarrow -\infty} \frac{1}{x}.$$

In fact

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$$

where n is an integer.

Key Idea – Limits of the Form $\frac{\infty}{\infty}$

Sometimes when we evaluate an infinite limit by direct substitution, we obtain an expression of the form

$$\frac{\infty}{\infty}$$

We refer to this as “indeterminant form”, which appears to be undefined.

If this is the case, then we need to

divide each term in the numerator and denominator by the highest power of x in the denominator.

Example:

Find $\lim_{x \rightarrow \infty} \frac{5x + 3}{4x - 2}$.

We can see that $5x + 3 \rightarrow \infty$ and $4x - 2 \rightarrow \infty$ as $x \rightarrow \infty$.

Since the fraction is in the form “ $\frac{\infty}{\infty}$ ”, we divide each terms in the numerator and the denominator by x .

$$\lim_{x \rightarrow \infty} \frac{5x + 3}{4x - 2} = \lim_{x \rightarrow \infty} \frac{\frac{5x}{x} + \frac{3}{x}}{\frac{4x}{x} - \frac{2}{x}} \quad (\text{dividing numerator and the denominator by } x)$$

$$= \lim_{x \rightarrow \infty} \frac{5 + \frac{3}{x}}{4 - \frac{2}{x}} \quad (\text{simplifying terms})$$

$$= \frac{5 + 0}{4 - 0} \quad (\text{using our special result})$$

$$= \frac{5}{4}$$

Exercise:

Evaluate:

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow \infty} \frac{x+1}{x-1} &= \lim_{x \rightarrow \infty} \frac{\frac{x}{x} + \frac{1}{x}}{\frac{x}{x} - \frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x}}{1 - \frac{1}{x}} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{x \rightarrow \infty} \frac{x^2 + 2x}{3x^3 + 4} &= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^3} + \frac{2x}{x^3}}{\frac{3x^3}{x^3} + \frac{4}{x^3}} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{2}{x^2}}{3 + \frac{4}{x^3}} \end{aligned}$$

$$= \frac{0+0}{3+0} = \frac{0}{3} = 0$$

$$\begin{aligned}
 \text{(c)} \quad \lim_{x \rightarrow \infty} \frac{x^3 - 2x}{3x^3 + 4} &= \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^3} - \frac{2x}{x^3}}{\frac{3x^3}{x^3} + \frac{4}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x^2}}{3 + \frac{4}{x^3}} \\
 &= \frac{1 - 0}{3 + 0} \longrightarrow = \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \lim_{x \rightarrow \infty} \frac{4x^5 - x^2}{2x^3 + 1} &= \lim_{x \rightarrow \infty} \frac{\frac{4x^5}{x^3} - \frac{x^2}{x^3}}{\frac{2x^3}{x^3} + \frac{1}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{4x^2 - \frac{1}{x}}{2 + \frac{1}{x^3}} \\
 &= \frac{\lim_{x \rightarrow \infty} 4x^2 - 0}{2 + 0} = \text{DNE}
 \end{aligned}$$